Project Report

VISION-SPEC QC

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1.Abstract

Industrial quality inspection is critical for ensuring product reliability, safety, and performance. Traditional manual inspection systems are slow, inconsistent, and prone to human error.

This project presents Vision Spec QC, a deep learning-based surface defect detection system that classifies industrial surface images into Pass and Defect categories.

Using transfer learning with MobileNetV2, the system achieves efficient training while maintaining high accuracy. The model attained a peak validation accuracy of 92.49%, demonstrating strong generalization capability and suitability for real-world industrial deployment.

2 .Introduction

Automated quality control is essential in modern manufacturing to maintain product integrity and reduce operational costs. Surface defect detection ensures products meet required standards and enhances customer satisfaction.

This project explores the application of Convolutional Neural Networks (CNNs) for surface defect detection, comparing:

- CNN built from scratch
- Transfer learning using MobileNetV2

The primary objective is to develop a reliable, scalable, and AI-driven inspection system.

3.Problem Statement

Manual inspection systems face several challenges:

- Human fatigue leading to inconsistency
- Subjective judgment errors
- Slow inspection speed
- High operational costs

Goal: Develop an automated deep learning system that:

- Improves inspection accuracy
- Reduces operational costs
- Increases inspection speed
- Enables scalable industrial deployment

4.Objectives

- Design a CNN-based binary classification model
- Implement transfer learning using MobileNetV2
- Achieve high validation accuracy
- Build a deployable and scalable inspection framework

5.Dataset Description

Dataset Used: Magnetic Tile Surface Defect Dataset
Original Classes:

- MT Blowhole
- MT Break
- MT Crack
- MT Fray
- MT Free (Normal)

Preprocessing for Practical Deployment:

- Combined all defect classes → Defect
- MT Free → Pass
- Image resizing: 128 × 128 pixels
- 80% training / 20% validation split
- Applied data augmentation for model robustness

6.Tools & Technologies

- Python
- TensorFlow
- Keras
- MobileNetV2
- Google Colab
- NumPy
- Matplotlib

6. System Architecture

Workflow:

1. Image Acquisition
2. Image Preprocessing
3. Feature Extraction
4. Classification Layer
5. Output Prediction

7. Model Development & Experimental Results

7.1 CNN From Scratch Architecture

Layers:

- Conv2D (32 filters, 3×3) + ReLU → MaxPooling (2×2)
- Conv2D (64 filters, 3×3) + ReLU → MaxPooling (2×2)
- Conv2D (128 filters, 3×3) + ReLU → MaxPooling (2×2)
- Flatten → Dense (128, ReLU) → Dropout (0.5) → Sigmoid Output

Parameters: 3,304,769

Training Configuration:

- Optimizer: Adam
- Loss Function: Binary Crossentropy
- Batch Size: 16
- Epochs: 5

Results:

Epoch	Train Accuracy	Validation Accuracy
1	0.6467	0.8728
2	0.7992	0.9017
3	0.8236	0.9249
4	0.8695	0.9191
5	0.8598	0.9075

Peak Validation Accuracy: 92.49%

8.2 Transfer Learning Using MobileNetV2

Architecture:

- MobileNetV2 (ImageNet Pretrained Weights)
- Global Average Pooling → Dense (128, ReLU) → Dropout (0.3) → Sigmoid Output

Parameters: 2,422,081

Results:

Epoch	Train Accuracy	Validation Accuracy
1	0.6994	0.9075
2	0.8538	0.8960
3	0.8509	0.9191
4	0.8779	0.9075
5	0.8867	0.9017

Peak Validation Accuracy: 91.91%

Observations:

- Faster convergence
- Stable validation accuracy
- Fewer trainable parameters
- Computationally efficient
- Scalable for industrial deployment

9.Comparative Analysis

Metric	CNN From Scratch	MobileNetV2
Peak Validation Accuracy	92.49%	91.91%
Training Time	Higher	Lower
Model Size	Larger	Smaller
Convergence	Slower	Faster % Stable
Industrial Deployment	Feasible	More Scalable

Conclusion: Scratch CNN achieved slightly higher accuracy, but MobileNetV2 is computationally efficient and scalable.

10.Model Evaluation & Analysis

9.1 Confusion Matrix

- Correctly identified 116 Defect samples
- Lower recall for Normal class due to class imbalance
- High sensitivity for defects
- Improvement needed: class balancing, advanced augmentation, threshold tuning

9.2 Output Heatmap Interpretation

- Heatmaps highlight defect regions (cracks/fractures)
- Enhances model explainability
- Useful for industrial validation and auditing

11. Conclusion

Vision Spec QC demonstrates that both scratch CNN and transfer learning are effective for surface defect detection.

- Scratch CNN: Higher peak accuracy (92.49%)
- MobileNetV2: Faster, stable, fewer parameters, easier deployment

The system is suitable for real-time industrial quality control.

12. Future Scope

- Grad-CAM for explainability
- TensorFlow Lite deployment
- OpenCV integration for real-time inspection
- Edge AI deployment on embedded systems
- Multi-class defect detection

13. References

- Howard et al., MobileNets, 2017
- Sandler et al., MobileNetV2, CVPR 2018
- Goodfellow et al., Deep Learning, MIT Press
- Krizhevsky et al., ImageNet Classification, NIPS 2012
- TensorFlow Documentation